
Comparative Analysis of Convolutional Neural Network and Transfer Learning Models for Brain MRI Tumor Classification

Abstract

Brain tumor classification from MRI scans, are crucial for early diagnosis and to gather proper preventive care. Manual interpretation by radiologists is both time consuming and also subject to error. This beckons the questions of exploring technology, and more specifically, automated deep learning approaches in classifying brain tumors. While AI architectures, such as MobileNetV2 and EfficientNetB0, have achieved and are known to have a strong performance recognizing everyday images like cars and animals, their effectiveness on specialized medical imaging tasks such as MRI images is not well established. The difference between the regular photos where these models excel in are that MRI scans differ fundamentally in texture, contrast, and structural patterns. This means that these models, which were originally trained on regular photos, may struggle to pick up on the visual details that distinguish one tumor from another.

In this work, we present a comparative study for a multiclass brain MRI tumor classification dataset across four different classes: glioma, meningioma, pituitary tumor, and no tumor. We created and evaluated a custom baseline CNN against MobileNetV2 and EfficientNetB0, both of which we also created, and additionally investigated the effect of data augmentation on transfer learning performance. Our experiments show that the baseline CNN achieves the highest test accuracy of 91.70%, outperforming both the MobileNetV2 (87.36%) and EfficientNetB0 (85.92%), with augmented MobileNetV2 yielding 86.81% suggesting that task-specific CNN architectures learn MRI features more effectively than frozen pretrained models on limited medical imaging datasets.

The link to the GitHub repository with all relevant code and information discussed in this paper is [here](#).

1 Introduction

Early and accurate brain tumor detection is critical for effective diagnosis and treatment planning. MRI scanning is one of the primary tools used for identifying brain abnormalities and masses, but manual interpretation by radiologists is needed to take further steps. Deep learning models, particularly Convolutional Neural Networks (CNNs), have achieved strong performance in the past when it comes to medical image classification tasks. However, these supervised models are often highly dependent on large, high-quality datasets to be trained and may struggle to generalize and produce proper results across specialized medical imaging data.

The recent success of transfer learning architectures pretrained on large datasets such as ImageNet offers a potential solution for improving medical image classification under limited amounts and types of data. For context, transfer learning architectures are neural network structures, pre-trained on massive datasets, that are altered for related tasks, and ImageNet is a visual database containing over

14 million annotated images designed for training and benchmarking computer vision models. Models such as MobileNetV2 and EfficientNetB0 are known to have strong feature extraction capabilities because of their extensive pretraining. The issue, however, lies as MRI scans differ significantly from natural images in terms of texture, contrast, and structural features. This is quite different than that of the datasets under ImageNet and thus may pose issues when it comes to correct classification. Furthermore, transfer learning models may require careful preprocessing and augmentation strategies to adapt successfully to medical imaging tasks.

In this work, we will address these challenges by creating and analyzing comparisons between deep learning pipelines for multiclass brain MRI tumor classification. For this, we will use a small-scale dataset containing four different tumor categories. This will include glioma, meningioma, pituitary, and no tumor classes. Then, we will use a CNN that we create, as a custom baseline, along with two transfer learning architectures: MobileNetV2 and EfficientNetB0. With these models, we will also experiment the impact of data augmentation on performance. What this means is we will rotate, dim/brighten, alter the data and see how the models respond with the training/testing data. Our experiments were conducted using a resource-efficient training pipeline and included preprocessing, normalization, augmentation, and evaluation using classification metrics and confusion matrices.

Our contributions are summarized as follows:

- We propose a comparative deep learning framework for evaluating CNN and transfer learning architectures on multiclass brain MRI tumor classification using limited medical imaging data.
- We conduct a systematic comparison between a custom CNN, MobileNetV2, and EfficientNetB0. Our results demonstrate that the baseline CNN achieves the highest test accuracy of 91.70%, outperforming the frozen transfer learning models.
- We validate the impact of data augmentation on transfer learning performance. Our analysis shows that augmentation produced comparable but slightly lower accuracy than the non-augmented MobileNetV2 model, suggesting that augmentation alone may not significantly improve generalization for this MRI dataset without deeper fine-tuning.

2 Related Work

2.1 Deep Learning for Brain MRI Tumor Classification

Traditional brain tumor classification approaches have relied heavily on CNNs due to their strong performance in medical image analysis tasks. Architectures such as AlexNet, VGG16, and ResNet have been widely adopted for medical image analysis tasks, with VGGNet known for its systematic use of small convolutional filters enabling efficient hierarchical feature extraction and representation learning (Simonyan & Zisserman, 2014; He et al., 2016). CNNs enable learning data-driven, highly representative, hierarchical image features from sufficient training data, and have been applied to medical image classification through strategies including training from scratch, using off-the-shelf pretrained features, and fine-tuning on target datasets (Shin et al., 2016). Past research has shown how well CNNs distinguish between tumors like glioma, meningioma, and pituitary growths in MRI scans. In one case, a transfer learning approach using VGG-16 hit a top accuracy of 98.69% (Rehman et al., 2020). Still, these supervised models are very dependent on having massive amounts of data. They also require significant amount of computing power (Gala, 2025). Another problem is that CNNs often get too used to their specific training data. Furthermore, CNNs tend to learn representations tightly coupled to their training dataset and distribution. This can result in a significant lack of performance under different conditions, with accuracy falling dramatically when models trained on controlled datasets are tested on “out-of-distribution” images (Mohanty et al., 2016).

2.2 Transfer Learning in Medical Imaging

The emergence and utilization of transfer learning has provided new opportunities for dealing with limited data availability in medical imaging scenarios. Pretrained architectures such as EfficientNetB0 leverage feature representations trained/learned from large-scale datasets such as ImageNet, allowing models to adapt more efficiently to downstream tasks with smaller datasets such as classifying brain tumors into glioma, meningioma, and pituitary categories using CE-MRI data (Vimala et al., 2023).

Recent studies have applied transfer learning using architectures such as EfficientNetB0, ResNet50, MobileNetV2, and VGG16 to detect and classify brain tumors from MRI images (Abiwinanda et al., 2022). Although they do seem to have good results, conventional transfer learning from pretrained models offer only partial benefits and frequently go through domain errors. This causes for conclusions to be made that there is a limited generalizability within medical imaging tasks (Guo et al., 2026). The findings from this seem to be that the direct application of these models may lead to suboptimal feature extraction and increased errors in classification within specific medical domains.

2.3 Data Augmentation and Generalization

Data augmentation is commonly used in the domain of medical imaging to improve model robustness and reduce overfitting. This is particularly helpful when datasets are limited in size. Common augmentation techniques applied to medical images include geometric transformations, photometric adjustments, and are essential because medical imaging datasets are often scarce due to very large amounts of needed annotations as well as limited data accessibility (Nalepa et al., 2019). These approaches increase dataset diversity by exposing models to varied data during training. This practice is often used to expand the availability of training data and to increase generalization, as deep learning models with large learning capacities often overfit to medical imaging datasets (Lee et al., 2021). Previous studies have shown that augmentation can improve performance in MRI classification tasks by encouraging models to learn invariant features specifically rather than memorizing training samples. It is important to note however, that applying a large number of augmentations does not necessarily lead to better performance by the models. Augmentation techniques must make sense and correlate to the type of analysis desired in order to positively influence model learning (Nalepa et al., 2019).

2.4 Transfer Learning Efficiency and Frozen Feature Extractors

Full fine-tuning of large pretrained architectures can be expensive and may require significant GPU resources that are not always available in academic or research settings. As a result, many transfer learning approaches utilize frozen features, where pretrained layers remain fixed while only lightweight classification layers are trained on the target dataset. Feature extraction uses fixed pretrained weights to extract relevant features for the target task, coupled with a task-specific classifier, and is known for its computational efficiency and potential for model reuse (Kolarik et al., 2024). Using frozen features offers computational efficiency and reduces overfitting risk, but may limit adaptability to new domains; fine-tuning allows the model to adjust its representations to domain-specific characteristics and typically yields superior performance, particularly when moderate amounts of labeled data are available (Cherti & Jitsev, 2021). However, frozen ImageNet-pretrained features may not adapt sufficiently to specialized MRI image patterns, limiting their effectiveness for medical classification tasks. Attempting to bridge the knowledge gap between natural and medical images by merely updating a small subset of pretrained parameters within a frozen model may not be sufficient to address the significant differences between these two domains (Xiao et al., 2024). This work investigates the tradeoff between lightweight frozen transfer learning models and a task-specific CNN trained directly on MRI images, while also analyzing the impact of augmentation on transfer learning generalization performance.

3 Methodology

Given the constraints of medical imaging variability and hardware, we designed a resource-efficient deep learning pipeline for multiclass brain MRI tumor classification. Our approach focuses on comparing a task-specific baseline CNN against lightweight transfer learning architectures while maintaining stable training and efficient computation. Additionally, we investigate the effect of data augmentation on transfer learning generalization performance in MRI classification settings.

3.1 Dataset and Preprocessing Pipeline

We utilize a publicly available Kaggle Brain MRI dataset containing 1200 MRI images across four classification categories: glioma, meningioma, pituitary tumor, and no tumor. The dataset is organized

into separate training and testing folders, with MRI scans stored according to their corresponding class labels.

To standardize the input representation across all models, MRI images are resized to 224×224 pixels. This resolution was selected to match the expected input dimensions of the standard transfer learning architectures while maintaining computational efficiency. Images used for the baseline CNN are normalized from the original pixel range $[0, 255]$ to $[0, 1]$ using simple rescaling. For MobileNetV2, preprocessing is performed using the architecture-specific `preprocess_input` transformation to align with ImageNet pretraining requirements.

The original training set is further divided into training and validation subsets using an 80/20 split with a fixed random seed to ensure reproducibility. During preprocessing, corrupted MRI image files producing local file read timeout errors were identified and removed to ensure stable model training and dataset integrity.

3.2 Baseline Convolutional Neural Network (CNN) Architecture

As a task-specific reference model, we implement a custom Convolutional Neural Network (CNN) trained directly on MRI images. Unlike transfer learning approaches that rely on pretrained ImageNet features, the baseline CNN learns feature representations entirely from the target MRI dataset.

The architecture consists of multiple convolutional blocks containing Conv2D and MaxPooling2D layers followed by fully connected dense layers and dropout regularization. Convolutional layers progressively learn spatial features such as tumor boundaries, texture variations, and structural MRI patterns, while max pooling reduces spatial dimensionality and computational complexity.

The final classification layer utilizes a softmax activation function to predict probabilities across the four tumor classes. Dropout regularization is applied to reduce overfitting during training.

3.3 Transfer Learning Architectures

A core contribution of this work is the comparative analysis between a task-specific CNN and lightweight transfer learning models. We investigate two pretrained architectures: MobileNetV2 and EfficientNetB0.

1. MobileNetV2

We utilize MobileNetV2 as a lightweight transfer learning model due to its efficiency and reduced computational complexity. The model is initialized using ImageNet-pretrained weights with the original classification head removed (`include_top=False`). To preserve the pretrained visual representations learned from large-scale natural image datasets while adapting the model to MRI classification, the pretrained backbone is frozen during training:

```
base_model.trainable = False
```

A lightweight classification head consisting of global average pooling, dropout regularization, and a dense softmax layer is added for tumor classification.

2. EfficientNetB0

We additionally evaluate EfficientNetB0, which balances network depth, width, and resolution using compound scaling. Similar to MobileNetV2, the pretrained EfficientNetB0 backbone is frozen while a lightweight classification layer is trained on MRI images.

This transfer learning strategy reduces computational requirements while enabling comparison between pretrained feature extraction and task-specific CNN learning.

To improve generalization performance and mitigate overfitting, we implement an online data augmentation pipeline during training. The augmentation strategy applies random geometric and photometric transformations to training MRI images while preserving the underlying anatomical structure.

3.4 Data Augmentation Pipeline

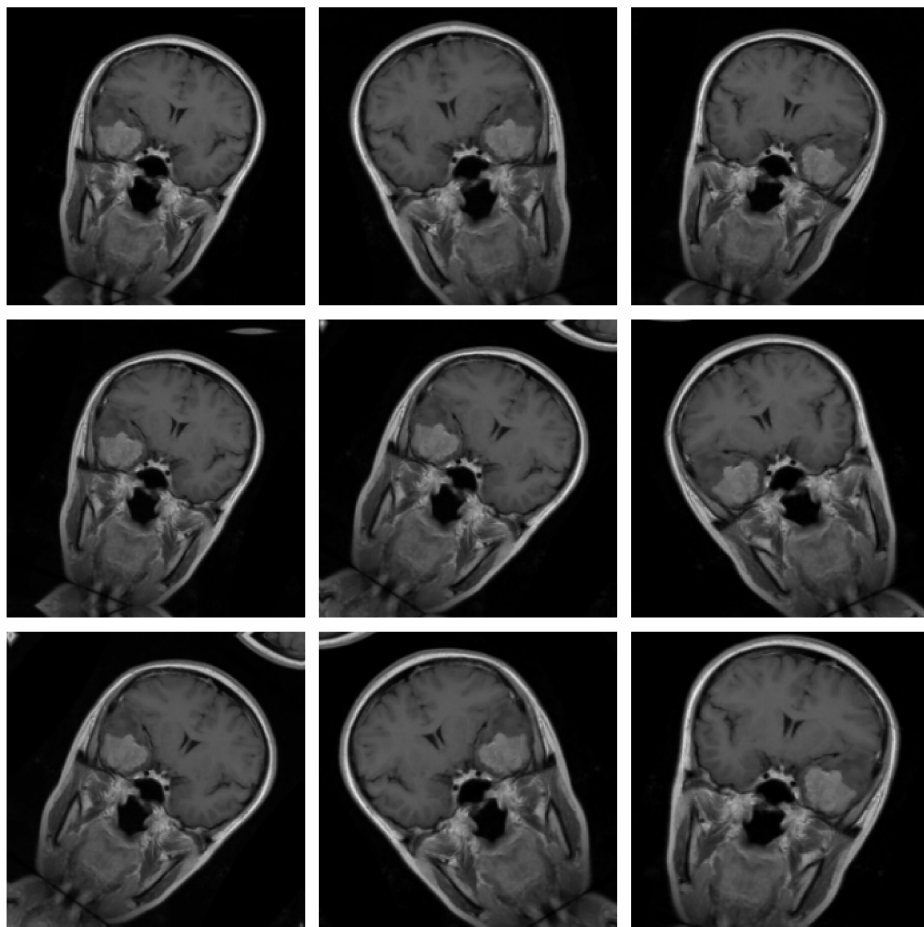


Figure 1: Visualization of augmentation samples applied during training. Random geometric and photometric transformations including flipping, rotation, zoom, and contrast adjustment were applied to data

1. Geometric Transformations To improve orientation invariance, we apply random horizontal flipping and rotation transformations. Random rotations within a small range are introduced to simulate variations in MRI positioning while avoiding excessive anatomical distortion.

2. Scale and Spatial Variations Random zoom transformations are applied to simulate variations in tumor scale and MRI cropping. These augmentations encourage the model to learn scale-invariant tumor features rather than memorizing fixed spatial patterns.

3. Photometric Transformations To improve robustness against intensity variations, random contrast adjustments are applied during training. These transformations simulate differences in MRI brightness and scanning conditions while preserving medically relevant structures. We additionally discovered that correct ordering of augmentation and MobileNet preprocessing operations was necessary for stable transfer learning performance. Applying augmentation after MobileNet preprocessing produced unstable training behavior and severely degraded accuracy, highlighting the importance of preprocessing consistency in transfer learning pipelines.

3.5 Training Configuration and Optimization

All models are trained using the Adam optimizer built into TensorFlow/Keras. We use sparse categorical loss for multiclass classification:

$$L = \text{SparseCategoricalCrossentropy}(y_{\text{true}}, y_{\text{pred}})$$

Evaluation Metrics

Training is performed for 10 epochs using mini-batch gradient descent. Due to local file-loading and memory constraints encountered during experimentation, smaller batch sizes were used to maintain stable training on consumer-grade hardware. The baseline CNN utilizes standard pixel normalization, while transfer learning models employ architecture-specific preprocessing pipelines. Model performance is evaluated using validation accuracy during training and final testing accuracy on unseen MRI images.

We evaluate model performance primarily using testing accuracy for final model comparison across the baseline CNN, MobileNetV2, and EfficientNetB0 architectures. During training, validation accuracy and validation loss are monitored to analyze convergence behavior and detect potential overfitting. In addition to overall testing accuracy, precision, recall, F1-score, and confusion matrices are used for class-level analysis of the baseline CNN model. Precision and recall provide insight into false positive and false negative behavior across tumor classes, while F1-score balances both metrics and is useful for evaluating multiclass medical image classification performance. Confusion matrices are additionally used to visualize class-specific prediction behavior and identify common tumor misclassification patterns.

4 Experiments & Results

4.1 Experimental Setup

We evaluate all models using the Kaggle Brain MRI dataset containing 1,200 MRI images across four classification categories: glioma, meningioma, pituitary tumor, and no tumor. The dataset was organized into separate training and testing folders, while the training data was further divided into training and validation subsets using an 80/20 split. All MRI scans were resized to 224×224 pixels to maintain compatibility with the transfer learning architectures and ensure computational efficiency.

All experiments were implemented using TensorFlow and Keras within a Jupyter Notebook environment. The models were trained for 10 epochs using the Adam optimizer with a learning rate of 1×10^{-4} . Due to local hardware and file-loading constraints encountered during experimentation, smaller batch sizes were utilized to improve training stability and memory efficiency.

During preprocessing, corrupted MRI image files producing local file read timeout errors were identified and removed to ensure stable training and evaluation. For transfer learning experiments, pretrained ImageNet feature extractors were frozen while lightweight classification heads were trained on the MRI dataset.

4.2 Baseline CNN Performance

We first evaluate the task-specific baseline CNN trained directly on MRI images without pretrained ImageNet feature extraction. The model achieved strong classification performance with a final testing accuracy of 91.70%, outperforming all evaluated transfer learning architectures.

The training and validation accuracy curves are shown in Figure 2, while the corresponding loss curves are presented in Figure 3. The model demonstrates stable convergence with relatively limited overfitting throughout training, indicating that the CNN successfully learned MRI-specific spatial and structural features directly from the dataset.

These results suggest that a task-specific CNN can effectively learn domain-relevant MRI representations without relying on pretrained natural-image features.

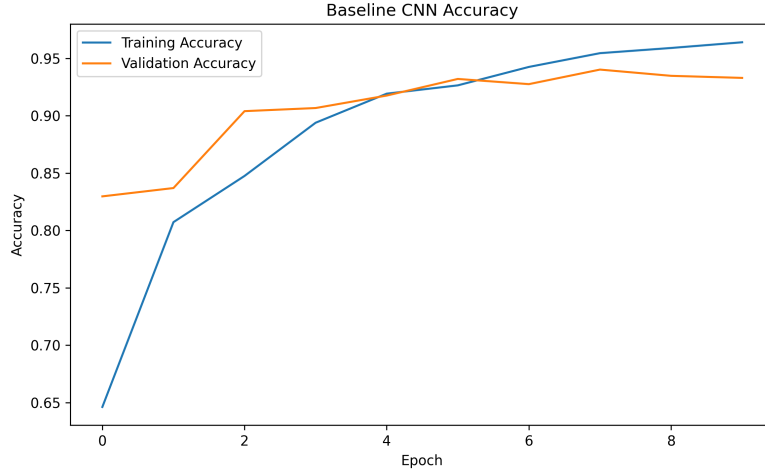


Figure 2: Training and validation accuracy curves for the baseline CNN model. The model demonstrates stable convergence and strong validation performance across training epochs.

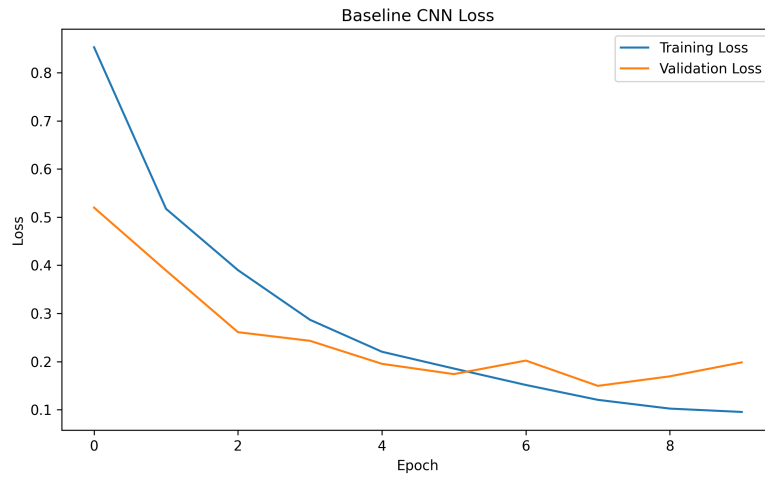


Figure 3: Training and validation loss curves for the baseline CNN model. Loss steadily decreases throughout training with limited signs of severe overfitting.

4.3 Transfer Learning Model Comparison

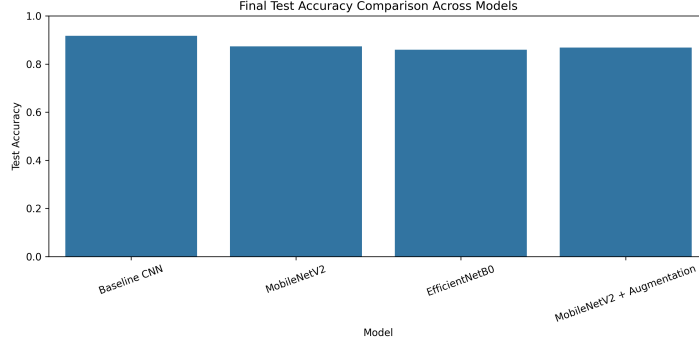
We next compare the baseline CNN against two lightweight transfer learning architectures: MobileNetV2 and EfficientNetB0. Comparisons across all evaluated models are summarized in Table 1 and Figure 4.

The baseline CNN achieved the highest testing accuracy of 91.70%, outperforming MobileNetV2 (87.36%) and EfficientNetB0 (85.92%). While transfer learning models benefited from pretrained ImageNet feature extraction, the frozen pretrained representations did not generalize as effectively to MRI image textures and structural patterns as the task-specific CNN.

One possible explanation is that MRI scans differ significantly from natural images commonly found in ImageNet, limiting the effectiveness of frozen feature extractors pretrained on everyday visual objects. These findings suggest that domain-specific feature learning may be more beneficial than frozen transfer learning representations for multiclass MRI tumor classification.

Table 1: Final testing accuracy comparison across evaluated models.

Model	Test Accuracy
Baseline CNN	91.70%
MobileNetV2	87.36%
EfficientNetB0	85.92%
MobileNetV2 + Augmentation	86.82%

**Figure 4:** Final test accuracy comparison across all evaluated models. The baseline CNN achieved the highest overall classification performance.

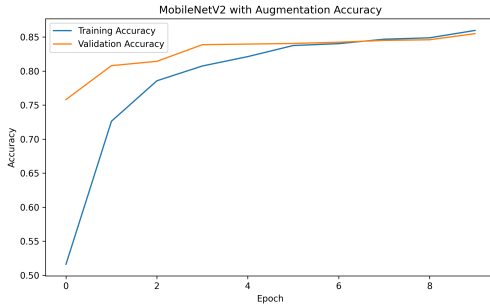
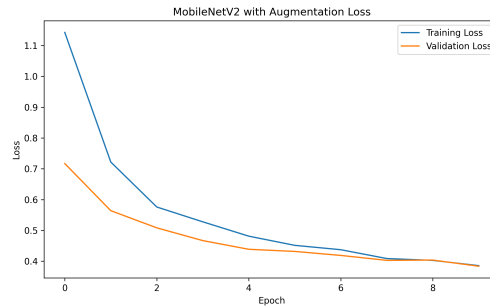
4.4 Impact of Data Augmentation

To analyze the effect of augmentation on transfer learning performance, we conducted an augmentation-based MobileNetV2 experiment using random flipping, rotation, zoom, and contrast transformations. Example augmentation outputs are shown in Figure 1.

The augmented MobileNetV2 model achieved a testing accuracy of 86.82%, producing slightly lower performance than the non-augmented MobileNetV2 model (87.36%). Although augmentation is commonly used to improve model robustness and reduce overfitting, the selected transformations did not significantly improve generalization performance for this MRI dataset.

One possible explanation is that MRI images already exhibit relatively standardized orientations and intensity distributions, limiting the usefulness of aggressive geometric augmentation. Additionally, because the transfer learning backbone remained frozen during training, the pretrained feature extractor may not have adapted effectively to the augmented MRI representations.

The training and validation accuracy curves for the augmented MobileNetV2 model are shown in Figure 5, while the corresponding loss curves are presented in Figure 6.

**Figure 4:** Training and validation accuracy curves for MobileNetV2 with augmentation. The augmented model achieved stable convergence but did not outperform the non-augmented configuration.**Figure 5:** Training and validation loss curves for MobileNetV2 with augmentation. Loss decreased consistently throughout training with moderate generalization performance.

4.5 Confusion Matrix and Classification Analysis

To further analyze classification behavior, we evaluate the baseline CNN using confusion matrices and class-level precision, recall, and F1-score metrics. The confusion matrix for the baseline CNN is shown in Figure 7.

The baseline CNN demonstrated particularly strong classification performance for pituitary tumor samples, correctly classifying the majority of pituitary MRI images with minimal misclassification into other tumor categories. The model also achieved relatively strong performance on meningioma samples, although a small number of predictions were incorrectly classified as glioma, no tumor, or pituitary classes.

Performance on glioma samples was comparatively weaker, likely due to lower class support and dataset imbalance following preprocessing and corrupted file removal. Additionally, the testing dataset contained no valid "no tumor" samples after preprocessing cleanup, limiting class-level evaluation reliability for that category and preventing meaningful performance analysis for no tumor classification.

Overall, the confusion matrix demonstrates that the baseline CNN achieved strong multiclass classification performance across the MRI dataset while highlighting the impact of class imbalance and dataset limitations on specific tumor categories.

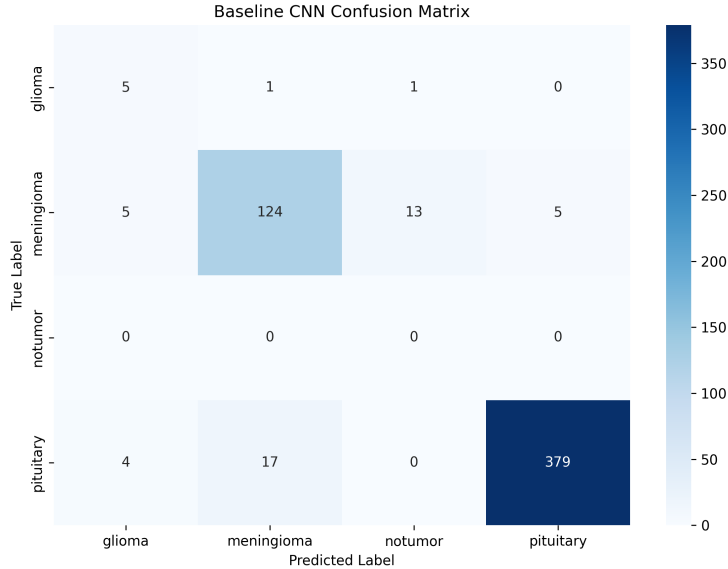


Figure 6: Confusion matrix for the baseline CNN model. The model demonstrates strong classification performance across most tumor categories, particularly for pituitary tumor classification.

5 Conclusion

We presented a comparative deep learning study for multiclass brain MRI tumor classification, evaluating a baseline CNN against MobileNetV2 and EfficientNetB0 with and without data augmentation. Experimental results, including accuracy comparisons, training curves, and confusion matrix analysis, demonstrated that the baseline CNN achieved the highest test accuracy at 91.70%, outperforming MobileNetV2 (87.36%) and EfficientNetB0 (85.92%). Additionally, the augmentation experiment showed that data augmentation produced comparable but slightly lower performance than the non-augmented MobileNetV2 model. These findings suggest that task-specific CNN architectures may better capture MRI-specific features than frozen ImageNet-pretrained models. Limitations include corrupted image files, class imbalance, and the use of frozen transfer learning backbones. Future work should explore larger balanced datasets, full fine-tuning strategies, and more advanced architectures to further improve MRI tumor classification performance.

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